Deepak Katukuri

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SAP SE
Recruiting Team – Student Careers Germany
Re: Working Student – Software Engineering, Requisition ID 457812

Dear SAP Recruiting Team,

I'm applying for the Working Student – Software Engineering position (Requisition ID 457812) on your Senior Engineering team. I bring nearly three years of production software engineering experience, a full-stack TypeScript/React skill set, and a working habit of testing new developer tools against a real benchmark before trusting them – which lines up closely with what this role asks for.

At ValueMomentum, I spent close to three years maintaining production P&C insurance applications on OneShield PAS: I migrated a full policy, claims, premium, and customer dataset onto the platform, and root-caused a recurring data-integrity defect in the claims pipeline that had been silently dropping records for a specific bond type. That work ran on Agile sprints with code review as a standing practice, not an afterthought.

On the "evaluate AI-native development practices" part of the role, the closest thing I can point to is my RAG pipeline project. I hand-built a hybrid retriever from scratch across four stages – BM25 and dense retrieval, RRF fusion, cross-encoder reranking – reaching 0.833 recall@5 on a 55-question labeled benchmark I wrote myself. I then rebuilt the identical retriever on LangChain, changing only the framework, and caught a silent 0.06-recall regression from LangChain's default BM25 tokenizer before it could hide in production – then fixed it to 0.854 recall@5. That's the workflow I'd bring to evaluating new AI-native tools here: build a benchmark first, adopt a framework only after it clears it, and write down what I learned.

On the TypeScript and React requirement: my production frontend experience is React-heavy – a full-stack health app I built on React, FastAPI, and PostgreSQL, with a rule-based recommendation engine and a Recharts analytics dashboard. To close the TypeScript side of that directly, I also built and shipped a Next.js 15 portfolio site in TypeScript with strict `tsc --noEmit` checking. That's the project I'd bring to an interview to present and defend line by line, per your request for non-trivial code I've written myself.

I should name one gap upfront rather than let it surface as a surprise later: most of my TypeScript work is recent – the portfolio site – not years-deep the way my JavaScript/React history is. I'm confident defending it, but wanted to be direct about where the depth currently sits.

I'm a full-time MSc Data Science student at the University of Europe in Potsdam, available immediately as a working student and full-time after graduation. My GitHub is github.com/Deepak589 – RAG_pipeline and NextJs-Portfolio are both active and reflect the work described above directly.

Thank you for considering my application. I'd welcome the chance to talk through any of this in more detail.

Best regards,
Deepak Katukuri